

Intro + Methods + Results Draft 1

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1 Introduction

Over recent years, it has become increasingly clear that the scientific community is facing a replication crisis (e.g., Forbes et al., 2023). A growing number of studies have reported difficulties reproducing previously published findings. This problem is often attributed to publication bias – the tendency to only publish significant results – and questionable research practices (QRPs). However, even studies aimed at removing the incentive for QRPs have painted a rather pessimistic picture of reproducible science (Auspurg & Brüderl, 2024). Many-analyst studies, where multiple independent researchers answer the same research question using the same dataset, have shown considerable variation in analytic strategies and results (Bastiaansen et al., 2020; Breznau et al., 2022; Gould et al., 2025; Hoogeveen, Sarafoglou, Aczel, et al., 2023; Kowall et al., 2025; e.g., Silberzahn et al., 2018). Reported effect sizes often vary from negative to positive, and from significant to non-significant, all within the same study. This means that there is not only considerable variation in the results but also in the substantive conclusions drawn from them. At first glance, this seems an immediate threat to the reproducibility of science. In an ideal world, one might expect that a single dataset should yield a single, ‘true’ answer, raising the question whether effect size variation should always be interpreted as problematic.

However, interpreting this variation without understanding its origins may not be that straightforward. A frequently raised criticism of many-analyst projects is that the posed research questions are often too vague, leaving analysts uncertain about how to proceed with data analysis and which variables should be included in their statistical models (Hoogeveen, Sarafoglou, Van Elk, et al., 2023; Kummerfeld & Jones, 2023).

For example, @silberzahn_many_2018 asked analysts to investigate the effect of skin tone of soccer players on the likelihood to receive a red card. This question can be interpreted different ways; are we interested in a descriptive association between skin tone and the likelihood of receiving a red card, or are we interested in a causal effect of skin tone on this

likelihood? Depending on these different causal effects, different variables need to be included in the model.

researchers may choose to identify different causal effects, each of which has its merit, but each of which implies different estimands and corresponding identification strategies (reasonable counterfactuals; 15, 36). For example, one might be interested in (i) identifying the descriptive difference in the probability of receiving a red card by skin color, or one might be interested in (ii) identifying the causal direct effect of skin color (“racial bias”).

This uncertainty is particularly problematic in situations where researchers are implicitly interested in finding causal relationships, which is often the case in many-analyst projects (Bulbulia, 2023; Del Giudice & Gangestad, 2021; Rohrer et al., 2025). In this context, including different confounders may correspond to different causal assumptions and thus different – potentially opposing – estimated effects. Therefore, when analysts select different variables, they may in fact be answering different questions (Simonsohn et al., 2015). From this perspective, effect size variation may indicate meaningful substantive differences rather than problematic analytical noise. On the other hand, if the effect size variation stems from arbitrary, equally justifiable decisions whose options do not reflect different substantive meanings, this may be considered more problematic (Del Giudice & Gangestad, 2021).

1.1 The AB-framework

For better interpretation of effect size variation in many-analyst projects, Auspurg & Brüderl (2024) have introduced a theoretical framework (hereafter referred to as the AB-framework) to distinguish between three different sources of variation, two of which are considered ‘harmful’ and one ‘helpful’ (see Table 1).

Table 1: Sources of variation

Harmful	Helpful
(a) Model uncertainty, including subjective analytical choices in data handling (e.g., missing data decisions, weighting etc.) and the selection of statistical models with different assumptions	(c) Substantively meaningful variation arising from different definitions of dependent, independent, and confounding variables, as well as differences in target populations (e.g., exclusion criteria or changes over time)
(b) Coding errors	

First, effect size variation may arise from model uncertainty (a), for example in missing-data handling or modelling assumptions, reflecting ambiguity about the correct approach to estimate the parameter of interest and its underlying data-generating process. For example, if different missing data handling strategies (e.g., listwise deletion versus multiple imputation) result in different mean effect sizes, while the underlying observed data remain the same, this would imply that the outcome depends on an arbitrary analytic choice rather than on the data

themselves. Secondly, effect size variation may be the result of sloppiness or hidden coding errors (b) made by the analysts. Both sources can be considered harmful, as they can introduce undesirable uncertainty into scientific analyses, making conclusions less credible.

In contrast, the third source concerns analytical choices that provide information about substantive variation that actually exists in the population (c). As mentioned earlier, selecting different target populations or confounders can lead to slightly different questions being answered. For example, if analyst X uses the full dataset, while analyst Y analyses only the men in that same dataset, X might find no effect in the general population, while Y finds that there is a male-specific positive effect. Given that the two analysts answered different questions, this source can be considered as helpful for the advancement of theories. By taking a closer look at the decisions made by analysts and the implications thereof, the AB-framework provides tools to interpret effect size variation in many-analyst projects.

1.2 Identifying sources of variation in many-analyst projects

Although several many-analyst studies have aimed to identify the sources of effect size variation, so far no consensus has been reached (Bastiaansen et al., 2020; Breznau et al., 2022; Gould et al., 2025; Hoogeveen, Sarafoglou, Aczel, et al., 2023; Kowall et al., 2025; e.g., Silberzahn et al., 2018). For example, Hoogeveen, Sarafoglou, Aczel, et al. (2023) found that the operationalisations of variables, rather than model choice, influenced the results the most. In contrast, Gould et al. (2025) showed that neither variation in variable selection nor model structure meaningfully predicted variability, but did not explicitly name another explanation source of variation. A similar pattern was found by Kowall et al. (2025), where data structure (cross-sectional versus longitudinal), rather than model or variable choice, likely had the largest impact on the results. Different studies thus highlight different sources of variation.

Given the wide variety of topics, designs, and variables across projects, it may be unrealistic to expect a single, fixed set of explanatory factors that applies to all studies. However, even identifying a study-specific source of variation might not be possible based on many-analyst studies alone. A key limitation of these projects is that the involved analysts make decisions in an uncontrolled way (Kummerfeld & Jones, 2023). This leads to the problem of sparse data, where little data is generally available for each individual decision, as unique combinations of choices are often chosen only once. This sparsity makes it difficult to properly disentangle individual sources of variation and provide systematic quantitative evidence of their relative influence. Consequently, solely looking at analytical decisions in many-analyst projects may not provide enough information to interpret effect size variation with the AB-framework.

1.3 Multiverse analysis

One approach to addressing the problem of data sparseness is multiverse analysis (Hoogeveen, Sarafoglou, Van Elk, et al., 2023; e.g., Steegen et al., 2016). First, this method identifies all

reasonable decisions, each with a specific number of options, and systematically combines them into a decision space (Liu et al., 2023). Since not all combinations may yield possible analyses, the set of valid analysis paths (universes) is always smaller than or equal to the full product. Then, results are reported across all valid universes. This way, multiverse analysis allows for the exploration of the entire decision space, rather than just a subset of choices made by analysts. As a result, substantially more data are available for each decision, enabling quantification of their specific impact on the results.

Although multiverse analysis seems a promising way of systematically evaluating the sensitivity of decisions, one of its limitations is that the decision space is often defined by a single researcher (Simonsohn et al., 2015). Consequently, some reasonable analysis paths may be overlooked. However, by basing the multiverse analysis on an existing many-analyst project, this problem can be solved. The decision space will then be determined by the “wisdom of the crowd” and represent realistic research choices (Auspurg & Brüderl, 2024).

Another limitation is that different universes with contradicting substantial implications may be included within the same multiverse. Del Giudice & Gangestad (2021) argue that as a solution, a multiverse should only include truly arbitrary, but not theoretically meaningful decisions, as the latter may lead to misleading results. However, by including both types of decisions and subsequently classifying them with the AB-framework, results that might otherwise be considered misleading could instead be interpreted in a meaningful way.

1.4 Present study

To assess the current state of the replication crisis using many-analyst projects, it is highly relevant to identify the most prominent sources of variation and their implications based on the AB-framework. Due to the descriptive nature of many-analyst studies, no consensus has yet been reached about what factors drive variability in the results, leading to the research question:

What is the relative contribution of harmful versus helpful analytical decisions in many-analyst projects to effect size variation when they are systematically evaluated in a multiverse?

Aiming to gain quantitative evidence of the relative influence of different sources of variation, the present study will conduct a multiverse analysis based on the many-analyst project by Kowall et al. (2025), where 16 teams analysed whether marital status (cohabiting married versus never married) predicts cardiovascular disease. This was chosen as a case study, because there was a large variety of both harmful and helpful decisions, providing a rich decision space to evaluate the impact of a wide range of different decisions. The decision space consists of six analytical decisions: handling missing data, choice of exclusion criteria, statistical model and definitions of dependent variables, confounders and target population. Before running the analysis, these decisions are classified as harmful/helpful using the AB-framework. The multiverse analysis is largely based on sampling and approximation methods proposed by Liu et al. (2023), where the decision space is sampled to calculate the decision sensitivity. Lastly, a

specification curve will be plotted to visualise how the outcomes relate to different analysis paths. For more details, see the Methods section.

-Our study has some limitations. First, the number of analysts was small. (...) Fourth, having the same effect estimate from all analyses would have been favorable for comparability. However, we did not restrict the choice of regression models because this is an important degree of freedom in data analysis. Therefore, we accepted different summary measures (e.g. odds ratios and hazard ratios) in the analyses. Nevertheless, as the disease risk was not very high (i.e. < 20%) the deviation between the figures for the different effect estimates may not be large. (Kowall et al., 2025)

2 Methods

2.1 Ethics

Ethical approval was granted by the Ethics Review Board of the Faculty of Social and Behavioural Sciences of Utrecht University.

2.2 Data

Consistent with Kowall et al. (2025), data from waves 1-7 (excluding wave 3) of the Survey of Health, Ageing and Retirement in Europe (SHARE) were used. SHARE is a longitudinal panel study conducted across 28 European countries, collecting data on health, social, economic and environmental policies, since 2004.

All variables needed for the construction of the decision space (outcome, exposure, adjustment, exclusion) were selected from different datasets that were available per wave. Then, one cleaned data set was created per wave, where the selected variables were linked through an ID variable. Lastly, all wave data sets were combined based on the selected Data Type option in the multiverse analysis. After data cleaning, but before applying the exclusion criteria and missing data handling, the samples consisted of:

- cross-sectional all waves: 138,807 participants across 336,804 observations.
- cross-sectional wave 1: 30,416 participants across 30,416 observations.
- longitudinal no strict baseline: 87,029 participants across 285,026 observations.
- longitudinal wave 1: 23,301 participants across 91,935 observations.

2.3 Definition of decision space

To define the decision space, all options of six different decisions made by the teams in Kowall et al. (2025) were identified based on information provided in the original publication (see Table 2). When systematically combining all options across decisions, the decision space consisted of 129,024 universes. After removing invalid research paths, 82,944 valid universes remained. Invalid paths included combinations of cross-sectional data with models requiring longitudinal data structure (Cox proportional hazards (Cox PH), discrete-time mixed-effects), cross-sectional wave 1 data with models requiring repeated measures (mixed-effects logistic regression, generalised estimating equations (GEE)), wave 1 data with outcomes that were not available at baseline (since last interview, cause of death), and Cox models with outcomes lacking time-to-event variables.

2.4 Classifying sources of variation using the AB-framework

The selected decisions were classified using the AB-framework. Missing data and statistical model choice were classified as harmful sources of variation. Operationalisation of the outcome variable, selection of confounders, and exclusion criteria were classified as helpful. Data structure was also classified as helpful, as the framework treats variations over time as representing different target populations (see Table 2).

2.5 Combining effect size measures

The statistical models yielded different effect size measures. Logistic regression, mixed-effects logistic regression, GEE, and discrete-time mixed-effects models produced odds ratios (OR). Cox PH models produced hazard ratios (HR). Log-binomial models produced risk ratios (RR) directly. To enable comparison across universes, all effect sizes were converted to RRs. Odds ratios and hazard ratios both approximate risk ratios when the outcome is rare, but can diverge substantially when the outcome is common (VanderWeele, 2020). Given that the overall prevalence in the unexposed group (p_0) was around 18% in all subsets of the data, conversions for both OR and HR were necessary.

Conversions to risk ratios often require knowing p_0 (Zhang & Yu, 1998). However, in a multiverse analysis, p_0 may vary across universes, depending on data type and exclusion decisions resulting in different data subsets. This could introduce arbitrary variation in the results, distorting the actual decision sensitivity.

Therefore, conversions proposed by VanderWeele (2020) were used, designed for scenarios when p_0 is unknown. For odds ratios, where the prevalence is expected to exceed 10%. (Huang et al., 2022a):

Table 2: Analytical decisions and their options based on the original many-analyst study, that were used in the present multiverse analysis.

Decision	Option	Notes / Deviations from original
Harmful sources of variation		
Missing Data	Complete cases (16) “No information” (1)	For complete cases, listwise deletion was applied. One team replaced missing values in categorical variables by “no information” to prevent listwise deletion. In the original, some analysts used imputed data provided by SHARE, these were not used in the present multiverse analysis.
Statistical Model	Cox proportional hazards model (6) Logistic regression (5) Log-binomial model (2) Mixed-effects logistic regression (1) Discrete-time mixed-effects model (1) Generalised estimating equations (1)	The original study reported different effect sizes (OR/HR/RR). For comparability across universes, in this study, all effect sizes were converted to risk ratios (RR).
Helpful sources of variation		
Data Type	Cross-sectional wave 1 (5) Cross-sectional all waves (1) Longitudinal, wave 1 as only baseline (3) Longitudinal, no strict baseline (10)	The cleaned datasets for each wave were combined in different ways, depending on the selected option.
Exclusion Criteria	Subjects married before 18 Subjects from Israel Subjects under 55 at recruitment Subjects born after 1954 Change in exposure (widowed/divorced)	The analysts in the original study often used idiosyncratic exclusion criteria that were used in the present study. Because the original paper did not specify whether combinations of these exclusion criteria were used, a maximum of one exclusion criterion per universe was applied. *Note: one team removed all participants from Israel, as this is not an European country.
Outcome	Heart attack / stroke ever (16) Age at heart attack / stroke (9) Heart attack / stroke since last interview (7) Main cause of death: heart attack / stroke (5)	Seven combinations of the dependent variables were used by the analysts in the original study. Although other combinations are possible, the present study only included the seven operationalisations from the original study.
Adjustment Set	Age (12) Sex (12) Country (7) Smoking (6) Education (4) Physical activity (4) No confounders (2)	To ensure feasibility of the present multiverse analysis, only adjustment variables used by at least four analysts, except ‘no confounders’, were included to create the adjustment sets.

Note. Numbers in parentheses indicate how many analysts selected each option in the original study.

Note. For the exposure variable, all teams but two used “cohabiting married” (CM). Some analysts also included registered partnership, and “married living apart”. However, since the original research question only focused on CM, this was the only variable used as exposure in the present study.

$$RR \approx \sqrt{OR} \quad (1)$$

For hazard ratios, where prevalence is estimated between 20% and 80% (Huang et al., 2022b; VanderWeele, 2020):

$$RR \approx \frac{1 - 0.5\sqrt{HR}}{1 - 0.5\sqrt{1/HR}} \quad (2)$$

Although the overall prevalence was slightly lower than 20%, a conversion not requiring a specific p_0 value was preferred to avoid introducing additional variation.

2.6 Round Robin sampling

Because the number of reasonable universes grows exponentially with the number of decisions and their options, the computational cost of multiverse analyses increases rapidly. To address this, Liu et al. (2023) proposed sampling universes rather than exploring the full decision space. They evaluated various algorithms and found that Round Robin sampling was one of the most effective methods for approximating multiverse results. Therefore, this method was applied in the present study. Round Robin sampling aims to represent each decision option at least once. The algorithm iteratively loops through all decisions and their options, sampling one universe at each stage without replacement, ensuring each sampled universe is included at most once. For a visual example of how the algorithm works, see Figure 1.

Although Liu et al. (2023) proposed sampling universes one by one and immediately processing the multiverse results, in this study universes were selected *a priori* before running the multiverse analysis. The sampler was implemented in two steps. First, decision options were mapped to corresponding row numbers in the full decision space. Then, row numbers were sampled according to the Round Robin algorithm and used to select universes from the original decision space. This approach was more computationally efficient than sampling directly from the decision space. A random seed was set to ensure reproducibility.

To ensure sufficient data were available for each option to quantify decision sensitivity, a minimum of 100 universes per option was sampled. As adjustment set had the most options (64), this resulted in 6,400 sampled universes in total.

Round Robin sampling algorithm

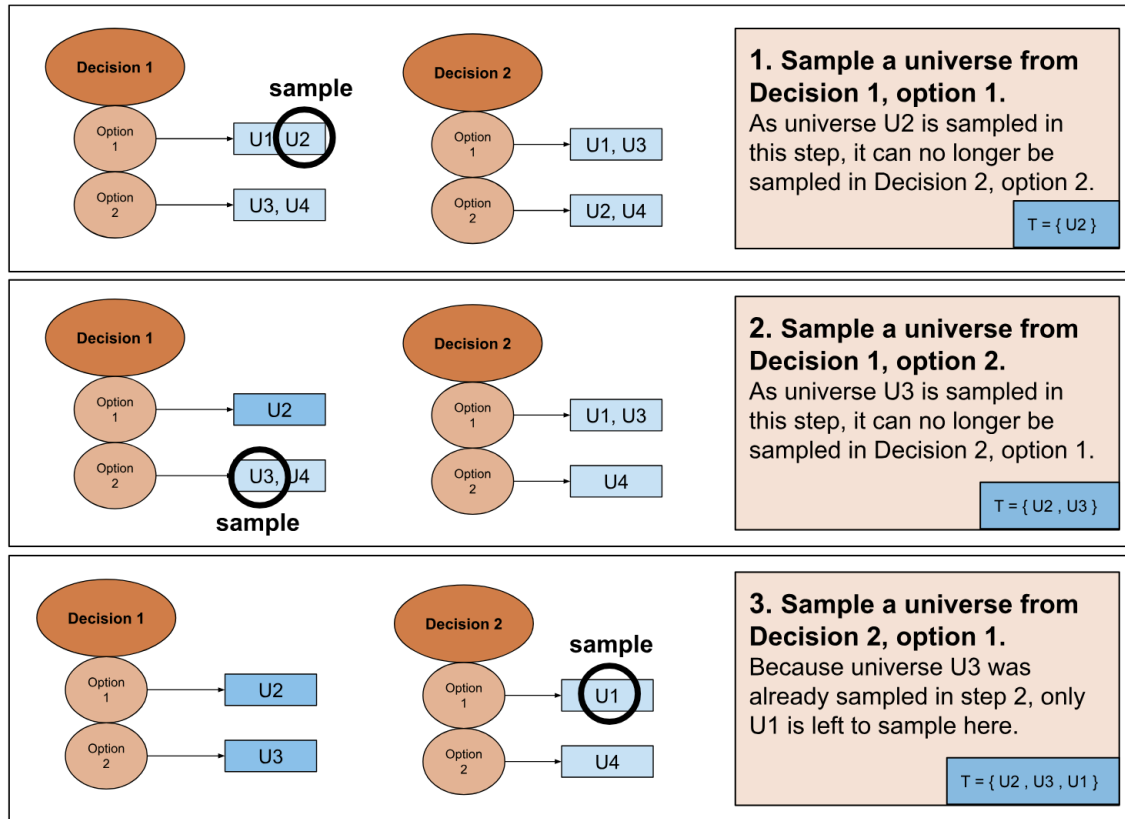


Figure 1: Schematic overview of Round Robin sampling algorithm. U: universe; T: the set of sampled universes. Figure was created by L. Ubbink.

2.7 Decision sensitivity

After converting all effect size estimates to risk ratios, density plots were used to visualise RR distributions across decision options. In addition, to visually assess the robustness of multiverse outcomes, each universe was plotted against its estimated effect size in a specification curve (Simonsohn et al., 2015).

How do specification curves work?

However, as both methods only provide descriptive information about the distribution of results rather than quantifying decision sensitivity, they were used as visual aids to inspect whether the outcome distribution was robust to different analytical specifications.

To quantify the decision sensitivity - the extent to which different options within each analytical decision lead to different outcomes - k -sample Anderson-Darling tests (AD tests) were used, as proposed by Liu et al. (2023). This test evaluates whether the RRs resulting from k options within a decision stem from the same empirical cumulative distribution function (ECDF) (Scholz & Zhu, 2019). If this null hypothesis was rejected, the decision was classified as sensitive. The AD statistic measures the weighted difference between the ECDFs of k samples, and is calculated as follows:

$$A_{kN}^2 = \frac{1}{N} \cdot \sum_{i=1}^k \frac{1}{n_i} \sum_{j=1}^{N-1} \frac{(N \cdot M_{ij} - j \cdot n_i)^2}{j(N-j)} \quad (3)$$

Where:

- N is the number of observations; the number of RRs resulting from all possible universes.
- k is the number of groups; number of options per decision.
- n_i is the number of observations per group; number of RRs resulting universes including a specific option.
- $Z_1 < \dots < Z_N$ is the pooled ordered sample; all RRs ordered from lowest to highest.
- j is the position in the pooled rank list Z_j .
- M_{ij} is the number of observations in a group that are smaller or equal to Z_j .
- $j(N-j)$ is used to add heavier weights to the tails of each distribution.

The standardised AD statistic (T.AD), which adjusts for number of groups and sample size, was used for between-decision comparisons. Higher T.AD values indicate more sensitive decisions. This statistic is calculated as follows:

$$T_{kN} = \frac{A_{kN}^2 - (k-1)}{\sigma_N} \quad (4)$$

For a more detailed calculation of σ_N , the variance of A_{kN}^2 , see Scholz & Zhu (2019) (Eq. 4).

Because there were 64 different adjustment sets, Kolmogorov-Smirnov tests (KS tests) were used for a more detailed evaluation of its decision sensitivity, comparing RR distributions for specifications including versus excluding each adjustment variable. The KS test statistic (D) represents the maximum absolute difference between the empirical cumulative distribution functions of the two samples, ranging from 0 (identical) to 1 (no overlap). Values closer to 0 indicate a higher likelihood that the two samples were drawn from the same distribution.

2.8 Supplementary Materials

All analyses were conducted in R (version 4.4.2, Team, 2024). All code scripts containing sampler and multiverse functions, and package versions documented in the lock file, can be found [here](#).

3 Results

3.1 Model convergence

Out of the 6,400 sampled universes, 1,487 failed to converge. Non-convergence mostly occurred in log-binomial models ($n = 1,102$), followed by generalised estimating equations (GEE; $n = 221$) and logistic regression ($n = 164$). The other three model types (Cox PH, mixed effects logistic, and discrete-time mixed effects) converged successfully in all cases. Across other analytical decisions, non-convergence was distributed approximately equally, suggesting these decisions did not substantially affect model convergence. In total, 4,913 universes yielded valid risk ratio estimates. The number of converged and non-converged universes per decision option can be found in Table 5.

3.2 Overall multiverse results

Across the 4,913 successful specifications, risk ratio estimates ranged from 0.71 to 1.68 (median = 0.95, mean = 0.94, SD = 0.1), with 79.6% yielding significant results ($p < .05$). Of all converged specifications, 65.3% showed a statistically significant negative association ($RR < 1$, $p < .05$), 14.3% a statistically significant positive association ($RR > 1$, $p < .05$), and 20.4% were non-significant. Within this sample of the full multiverse, marriage thus appeared both as a protective factor ($RR < 1$) and a risk factor ($RR > 1$) for cardiovascular events, depending on the analytical specification. Figure 2 shows the full distribution of the effect estimates across universes, arranged from smallest to largest risk ratio, with significance indicated by colour.

Several patterns emerged in the specification curve (Figure 2). Universes including age as an adjustment variable were clustered around higher RRs (mostly ≥ 1), while universes without age were clustered around lower RRs (all < 1), suggesting a potential effect of age on the relationship between marriage and cardiovascular events. Inclusion of sex as an adjustment variable showed more or less the opposite pattern: when it was included, RRs were clustered around lower values (almost all < 1), while exclusion of sex more often resulted in higher RRs. Not including any confounders always resulted in significantly negative results ($RR < 1$). The highest RRs (> 1.25) only occurred in specifications including Cox PH models, although almost none of these estimates reached significance. Log-binomial models, in contrast, yielded only RRs below 1, almost all of which were significant. Furthermore, effect estimates seemed robust to data type (except cross-sectional W1), missing data handling, outcomes (except when

combining “age at event” and “event ever”), and exclusion criteria, with no clear clustering patterns across these decisions.

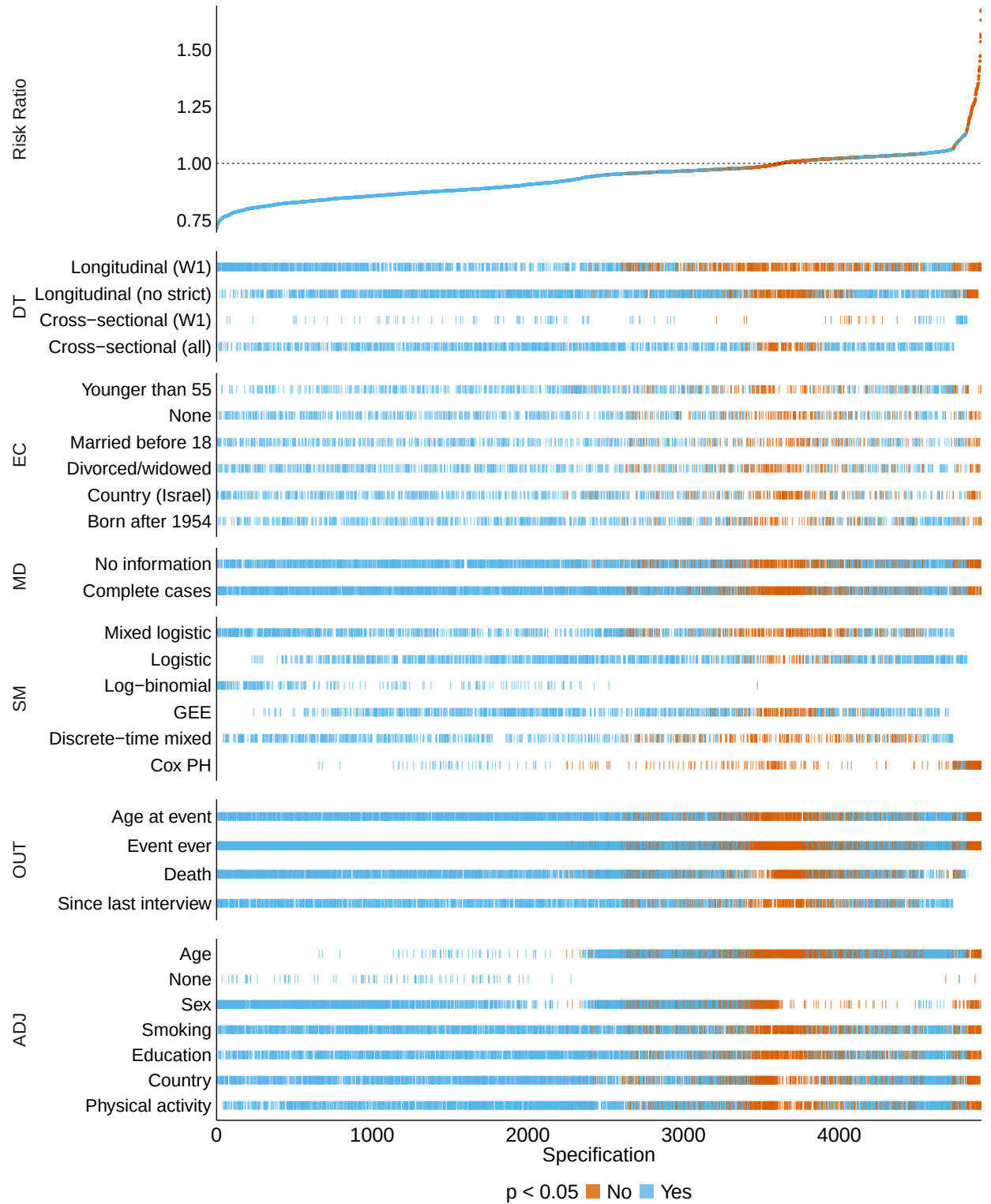


Figure 2: Specification curve multiverse results

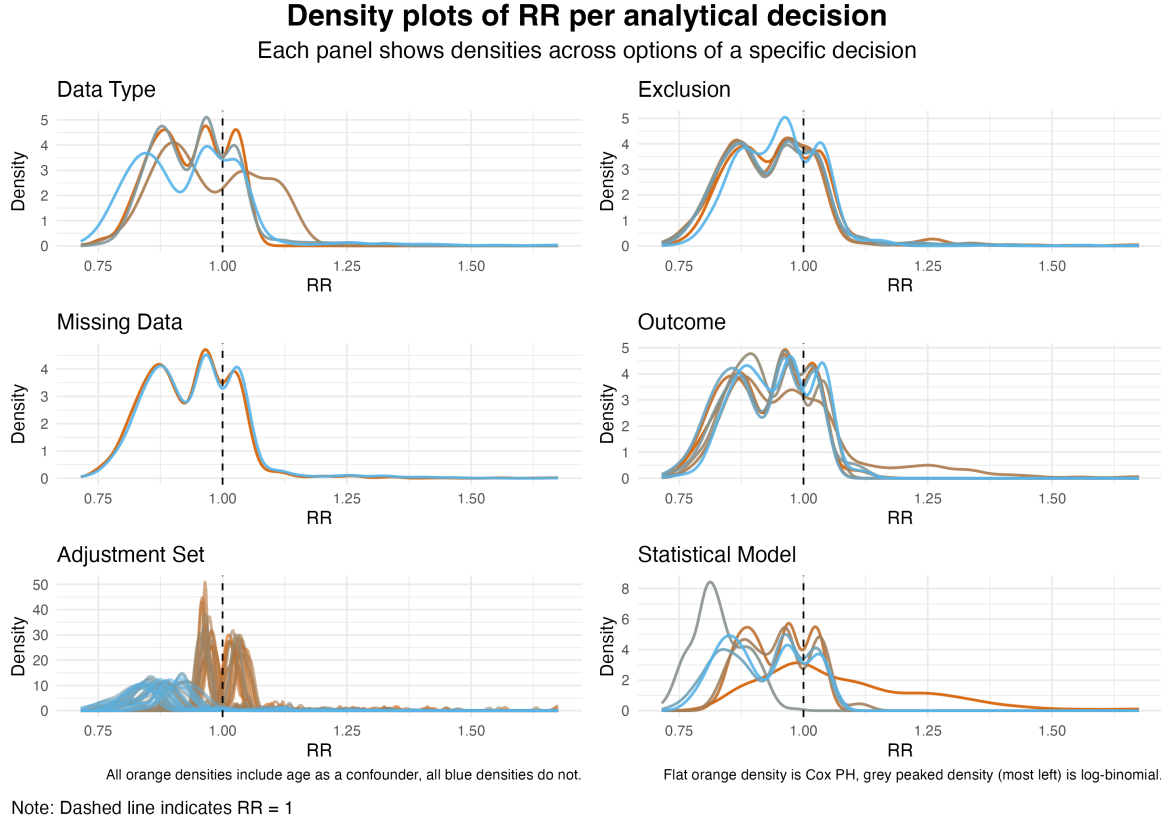


Figure 3: Densities of risk ratios (RR) across decisions and their options.

3.3 Decision sensitivity

To examine the effect size distributions across decision options in more detail, density plots were used (see Figure 3). Data type, missing data, exclusion criteria, and outcome showed largely overlapping RR distributions across their options. However, in case of statistical models and adjustment sets there were clear separations in RR distributions between options. Consistent with the pattern shown in Figure 2, log-binomial models were clustered below RR values of 1, while the density of Cox PH showed a long tail extending above 1. Similarly, specifications including age in the adjustment set showed a clear shift towards higher RRs compared to those without age as a confounder.

Mean RR ranges across options confirmed the patterns found in the specification curve and density plots (Table 5). Statistical model (0.83 - 1.08) and adjustment set (0.83 - 1.05) showed the largest variation in mean RR across options, while other decisions showed smaller ranges: exclusion criteria (0.94 - 0.94), missing data (0.93 - 0.95), data type (0.93 - 0.97), and outcome (0.92 - 0.98).

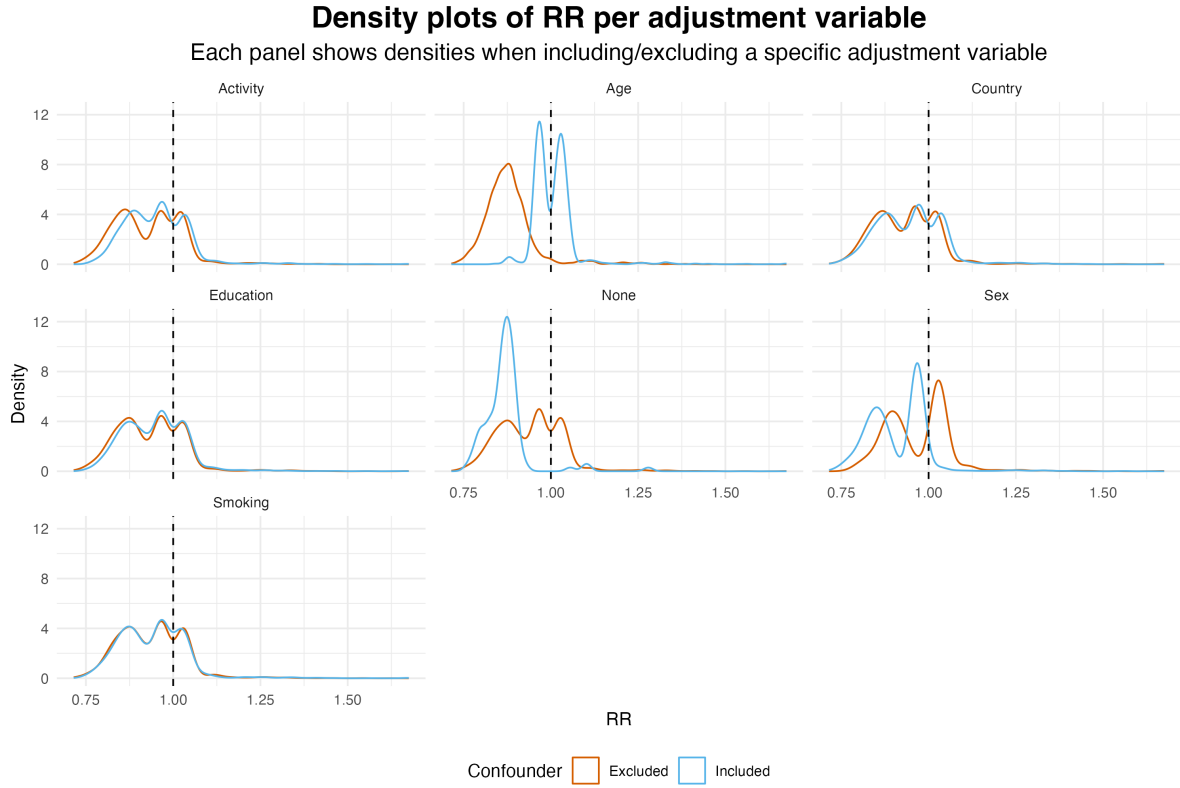
Table 3: K-sample Anderson-Darling tests for distributional differences across analytical decision options, with higher standardised AD values indicating more sensitive decisions.

Decision	k	T.AD	p-value
Missing Data	2	5.89	0.002
Exclusion Criteria	6	13.13	< .001
Data Type	4	50.88	< .001
Outcome	7	52.26	< .001
Statistical Model	6	307.10	< .001
Adjustment Set	64	394.79	< .001

Note. k = number of options; T.AD = standardised AD value.

A k-sample Anderson-Darling test (AD test) was used to quantify decision sensitivity (for the ECDF per decision see Figure 5). As shown in Table 3, AD tests for all decisions were highly significant (almost all $p < .001$), indicating distributional differences across options within each decision. Adjustment set and statistical model showed substantially higher T.AD values (307.1 and 394.79, respectively) than other decisions (missing data: 5.89; exclusion: 13.13; outcome: 50.88; data type: 52.26). This indicates that statistical model and adjustment set had the largest impact on the distribution of effect estimates. The number of options per decision did not seem to substantially affect the T.AD values.

Kolmogorov-Smirnov tests (KS test) were used to compare RR distributions for specifications including versus excluding each adjustment variable. For the ECDFs across adjustment variables, see Figure 6. As can be seen in Table 4, inclusion of all adjustment variables, except smoking, resulted in significantly different distributions than when these variables were excluded ($p < .001$). Age showed the largest effect (0.88), followed by not including any confounders (0.54) and sex (0.45). Physical activity, country, and education showed moderate, but significant, effects ($D_{activity} = 0.16$, $D_{country} = 0.1$, and $D_{education} = 0.08$). This confirms that the adjustment set sensitivity seems to be driven primarily by age, sex and the decision not to include confounders at all.



Note: Dashed line indicates $RR = 1$

Figure 4: Densities of risk ratios (RR) across inclusion/exclusion of different adjustment variables.

The adjustment set sensitivity driven by age, sex and none is also visible in the RR density distributions (Figure 4). Including age as an adjustment variable shifted the RR distribution to the right (higher RRs), while including sex shifted it to the left (lower RRs). The density pattern of specifications with no confounders resembled that of specifications excluding age. This suggests that adjustment set sensitivity is likely mainly driven by whether age is included at all, which is consistent with the clustered pattern observed in the specification curve (Figure 2). Other adjustment variables did not result in pronounced differences between RR distributions.

Table 4: Kolmogorov–Smirnov test for distributional differences across adjustment set options, when including or excluding specific adjustment variables. D: absolute max distance between the ECDFs of the two samples.

Adjustment variable	Included	Excluded	D	p-value
Smoking	2332	2581	0.03	0.341
Education	2214	2699	0.08	< .001
Country	2436	2477	0.10	< .001
Physical activity	2431	2482	0.16	< .001
Sex	2409	2504	0.45	< .001
None	100	4813	0.54	< .001
Age	2330	2583	0.88	< .001

Note. D = KS test statistic (range: 0 - 1), smaller numbers indicate smaller distributional differences.

3.4 Descriptive statistics SHARE dataset

In the SHARE data that was used for all analyses, married individuals were on average 4 years younger than unmarried individuals (65 versus 69 years old). Additionally, the distribution of sex differed between groups: 51% of married individuals were female, compared to 69% in the unmarried group.

4 Discussion

4.1 Summary of results and comparison with original study

The present multiverse analysis aimed to quantify the relative influence of different analytical decisions on effect size estimates for the association between marital status and cardiovascular events, based on the many-analyst project by Kowall et al. (2025). Across 4,913 specifications, risk ratio estimates ranged from 0.71 to 1.68, with 79.6% of the universes yielding significant results. This is broadly in line with the effect size range found in the original study, which was [OR: 0.72 to HR: 1.31]. After applying the RR conversions used in the present study, the original effect sizes would range from 0.85 to 1.21, indicating the original study captured a slightly more narrow RR range than the present multiverse. This is not surprising, as the original study only included 16 analyses. In both studies marriage was protective or harmful for cardiovascular health depending on the analytical specification. This underscores that reasonable specifications can lead to substantially different conclusions, consistent with findings

of earlier many-analyst projects (Bastiaansen et al., 2020; Breznau et al., 2022; Gould et al., 2025; Hoogeveen, Sarafoglou, Aczel, et al., 2023; e.g., Silberzahn et al., 2018).

The results of the present study diverge from the conclusions of the original study, which found that the cross-sectional versus longitudinal distinction was the most important source of variation in effect sizes, with statistical model and adjustment set selection playing only a minor role. An opposite pattern emerged in the present study. Although AD tests for all decisions were significant - indicating they were all sensitive - statistical model and adjustment set seemed to be the most prominent drivers of variation in RR distributions, while data type only resulted in minimal distributional differences. Based on a more detailed KS test of the adjustment set, age, sex, and the exclusion of confounders seemed to have impacted the sensitivity of this decision the most. The distribution of age and sex across groups could be a plausible substantive explanation for this finding. Unmarried individuals in this sample were older and predominantly female (69%), meaning that without adjustment for age or sex, the effect of marriage was affected in opposing directions: the younger average age of married individuals may have inflated the apparent protective effect of marriage, while the higher proportion of males in this group may have attenuated this effect, assuming men have a greater cardiovascular risk (Pana et al., 2024).

The finding that age and sex were likely the most important drivers of adjustment set sensitivity, might explain the discrepancy between the present and original study. In the original, 12 out of 16 researchers included both age and sex in their analyses, and the researchers also included these confounders in their additional sensitivity analyses, leaving little data available about the effect of not including these variables, potentially attenuating the sensitivity of this decision. On the other hand, this inconsistency in decision sensitivity between the original and the present study might reflect a fundamental difference between the two approaches. The original additional analyses varied one decision at a time under a fixed configuration, whilst a multiverse simultaneously crosses all decisions, capturing aggregate influence across the full space of specifications. These two quantities need not rank decisions in the same order. The finding that adjustment set sensitivity was driven primarily by age, sex, and the exclusion of confounders altogether suggests that confounder selection introduces meaningful variability that one-at-a-time sensitivity analyses may underestimate. Rather than contradicting the original study, these results extend it: when analyst degrees of freedom are varied jointly, modelling choices and confounder selection emerge as consequential sources of uncertainty that warrant explicit acknowledgement.

- *Table 1 shows the influence of different definitions of the exposure and the outcome on the effect estimates. For the ten variants used in the additional analyses 2, the relative risks ranged from 1.101 to 1.229.*
- Present range differed slightly from the effect size range in the original Kowall et al. (2025), which was [OR: 0.72 to HR: 1.31], converted: 0.85 to 1.21.
- *In the longitudinal analyses with wave 1 as baseline (N=5,449), the relative risks with the adjustment sets chosen by the participants ranged between 1.10 and 1.16 (additional*

analyses 3, Fig. 4).

-Five groups performed sensitivity analyses using different regression models: except for one analyst using generalized estimating equations (GEE model) in addition to a Cox model, the choice of the regression model had little influence on the effect estimate (Table 3). E.g., in one analysis, the OR was 0.97 (95% 0.84–1.11) when a discrete time mixed model was used, and the HR was 0.99 (95% CI: 0.86–1.13) when a Cox regression model was fitted.

-Four used age as a proxy for time on study. – Although more possible operationalisations were used by different analysts, most of them used age as a time variable so that was also done in this multiverse.

-The effect estimates (hazard ratios, odds ratios, or relative risks) for the association between marital status (cohabiting married versus never married) and cardiovascular events ranged from 0.72 to 1.31. The variation in effect estimates was smaller when the strictly longitudinal studies (and longitudinal analyses with wave 1 as the only baseline in particular) and the strictly or partly cross-sectional studies were considered separately. Almost each group of analysts used a different adjustment set for confounding, which had only a small effect on the range of effect estimates in our example data set (relative risks ranged from 1.10 to 1.16). Differences in the precise definition of exposure and outcome, and the choice of the regression models also had an only small influence on the effect estimate.

-The sensitivity analyses by some analysis groups and the additional analyses 2 and 3 suggest that researcher degrees of freedom like the choice of precise definitions of exposure and outcome, the choice of adjustment sets, and the choice of regression models each had only small impact on the effect estimates. Cross-sectional analyses, longitudinal analyses with only wave 1 as baseline, and longitudinal analyses including also participants recruited at later waves actually answer different questions.

-Results of analyses with comparable research questions differed less. For longitudinal analyses with wave 1 as the only baseline, the three hazard ratios were 1.12, 1.19 and 1.31. Among the cross-sectional analyses, analysis J which is a strictly cross-sectional analysis of wave 1 and analysis K which is a strictly cross-sectional analysis of all waves and which includes many participants more than once, are barely comparable to the remaining four partly cross-sectional analyses for which effect estimates range from 0.83 to 0.96. GEK: in mijn analyse zijn alle gemiddelde RRs van Data Type heel vergelijkbaar across data types! en allemaal gemiddeld rond 0.94/0.95 dus veel lager dan die van Kowall ?

- The additional analysis 1 clearly demonstrates that the different results from cross-sectional and longitudinal analyses do not result from confounding by other factors like definition of exposure and outcome, the choice of the adjustment set or the choice of the regression model. (...) In the present study, much of the variation can be attributed to the type of study on which the analyses were based (cross-sectional, longitudinal, hybrid), and to the correspondingly large differences in the size of the analysis data sets. (...) Our results show that many “classical” aspects of epidemiology (type of regression model, definition

of outcome or exposure, selection of confounders) played only a minor role, in contrast to the choice between cross-sectional and longitudinal analysis. In de originele studie zeggen ze eigenlijk dat vooral DT uitmaakt, en SM en ADJ niet echt, maar bij mij is dat juist wel zo, hoe kan dat?

→ age and sex were included by 12/16 analysis teams, so maybe that's why the effect sizes differ? als er bijna geen universes zijn waar deze ADJ niet in zitten zie je ook niet dat deze een duidelijk verschil teweeg brengen? en dan mis je dus miss die lagere RRs die met het niet-includen van age gepaard gaan?

4.2 Limitation

- It stood out that 1102 log-binomial models did not converge (only 232 did) - why? The high non-convergence rate in log-binomial models may indicate model misspecification, possibly due to predicted probabilities exceeding 1 — a known limitation of log-binomial models with certain data structures. [reference?]
- The fact that all cox models, even after conversion, resulted in very high RRs suggests that model assumptions fundamentally alter the conclusion. Especially because only 3 out of 7 outcome operationalisations and only two data types were compatible with Cox, just choosing a statistical model also means implicitly making decisions about other facets of the analysis. Keuzes staan niet individueel op zichzelf.
- All AD tests reached statistical significance, but this likely reflects the high power resulting from the 4,913 specifications rather than practically meaningful differences for all decisions. The T.AD values and descriptive ranges suggest that only statistical model and adjustment set meaningfully influenced results, while other decisions showed minimal practical impact despite statistically significant distributional differences.
- The age and sex differences between groups help explain the sensitivity to adjustment. Unmarried individuals were older and predominantly female. Without adjustment, married individuals appear to have lower risk of HA due to being younger inflating the apparent protective effect of marriage. At the same time married individuals appear to have higher risk of HA due to having more males in the group attenuates potential protective effects of marriage [reference: relationship sex/HA?] -The pattern for 'no confounders' aligns with age exclusion, suggesting age is the dominant confounder. This is in line with the D statistic. When there are no confounders included, the RRs are lower, while when any confounder is included, the RRs become higher.
- Causal inference literature: why in/excluding covariates matters.
- AB framework does not work
- AD test does not work - too sensitive

- What would be better than AD test?
- Limitation: very specific, not very generalisable, should be repeated with more datasets/many analysts.
- Limitation: HR conversion approximation maybe not optimal
- Sampled subset of full multiverse instead of full multiverse – aan de andere kant, grotere sample al helemaal overpowered?
- What should analysts do when model choice changes conclusions? And: make DAGs
Three used directed acyclic graphs (DAGs), but the DAGs also varied considerably. (...)
One group explained that the research question was not aiming at causal inference in their views and did not adjust for confounders at all. (Kowall et al., 2025)
- You can't determine which specification is "correct" even when they have different conclusions - but maybe that means people should always do multiverse to look at the outcome distributions?

5 References

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6 Appendix

Empirical cumulative distribution functions (ECDF) of RRs per analytical decision

Each panel shows ECDFs across decision options

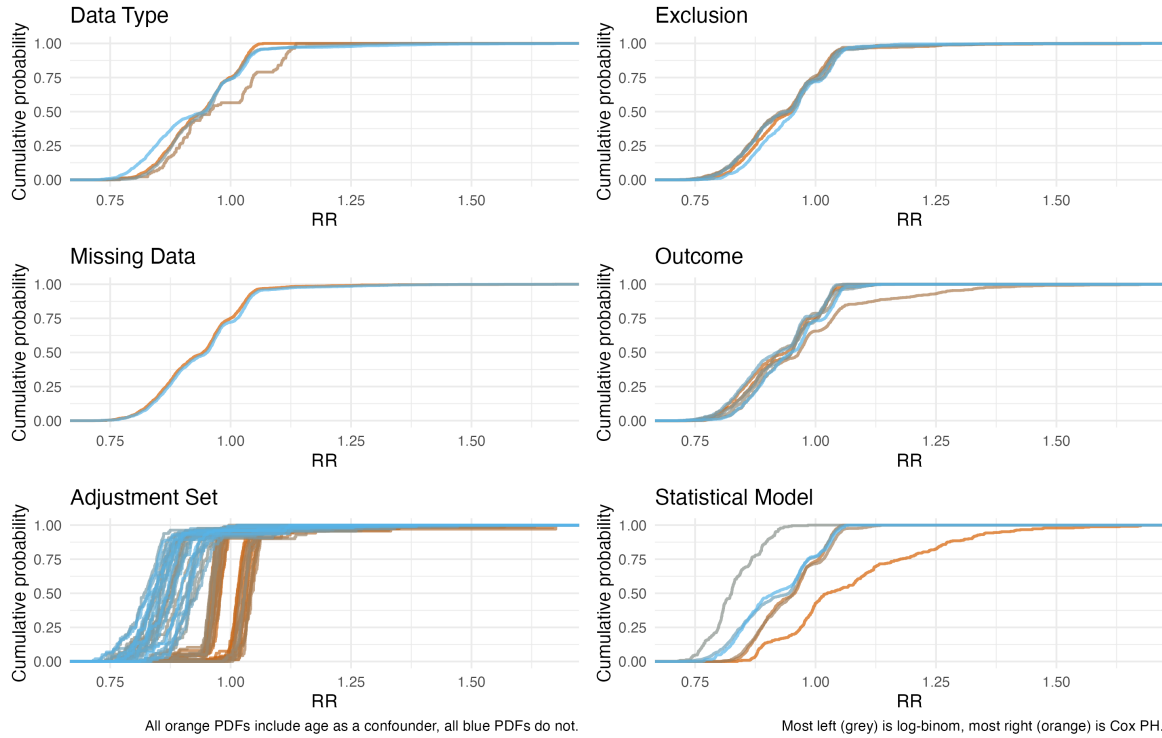


Figure 5: Empirical cumulative distribution functions of risk ratios (RR) across decisions and their options.

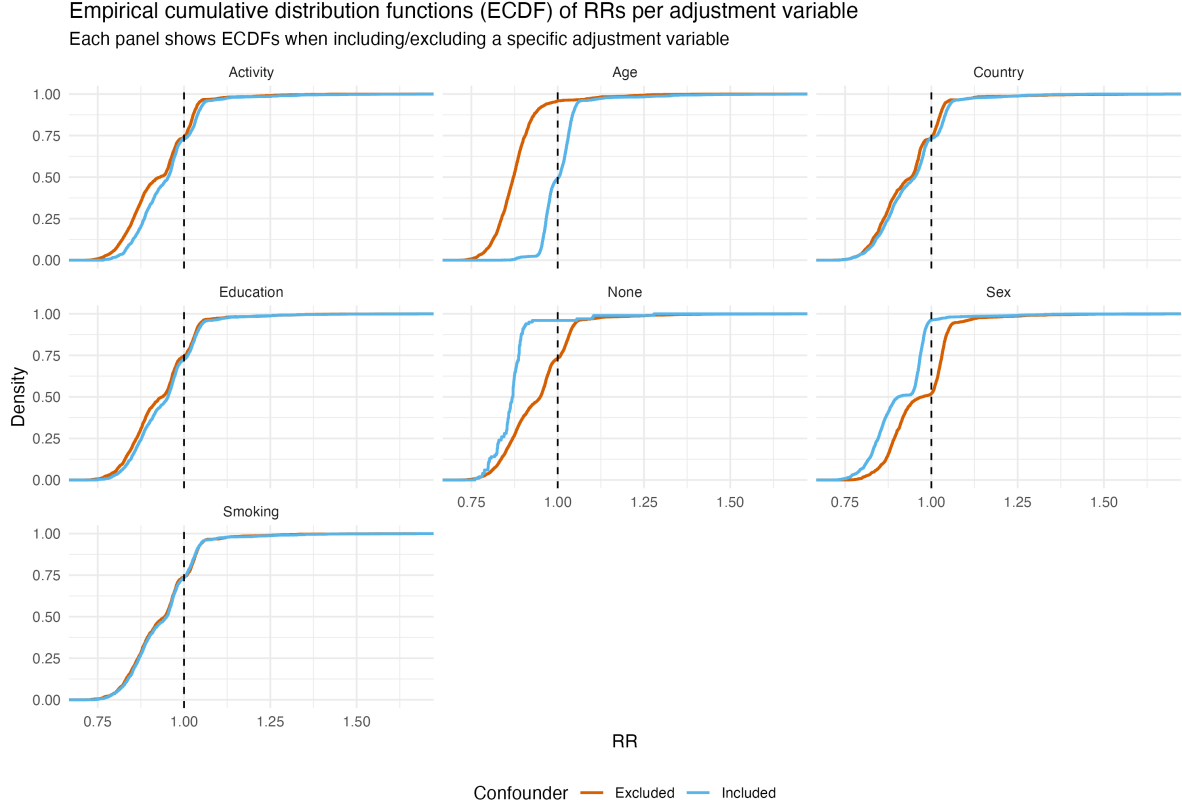


Figure 6: Empirical cumulative distribution functions of risk ratios (RR) across inclusion/exclusion of different adjustment variables.

Table 5: Number of universes including specific decision options, non-convergence, RRs, their range and percentage of significant RRs across decisions and their options.

Decision	Option	Successful Universes	Non- convergence	Mean RR	RR Range	% Sig
Statistical Model	Log-binomial	232	1102	0.83	[0.71, 0.98]	100%
Statistical Model	Mixed effects logistic regression	1278	0	0.92	[0.73, 1.07]	77%
Statistical Model	Discrete-time mixed effects	832	0	0.92	[0.76, 1.07]	74%
Statistical Model	GEE	1052	221	0.95	[0.8, 1.06]	85%
Statistical Model	Logistic regression	1188	164	0.95	[0.8, 1.14]	93%
Statistical Model	Cox PH	331	0	1.08	[0.84, 1.68]	24%
Data Type	Cross-sectional (all waves)	1237	403	0.93	[0.74, 1.07]	91%
Data Type	Longitudinal (wave 1)	1669	563	0.93	[0.71, 1.68]	69%
Data Type	Longitudinal (no strict baseline)	1869	411	0.95	[0.75, 1.35]	82%
Data Type	Cross-sectional (wave 1)	138	110	0.97	[0.77, 1.14]	86%
Missing Data	Complete cases	2584	637	0.94	[0.72, 1.68]	80%
Missing Data	No information	2329	850	0.94	[0.71, 1.68]	80%
Exclusion Criteria	Country	833	261	0.93	[0.73, 1.54]	76%
Exclusion Criteria	Divorced/widowed	819	277	0.93	[0.73, 1.47]	79%
Exclusion Criteria	Married before 18	778	248	0.94	[0.71, 1.45]	82%
Exclusion Criteria	None	804	263	0.94	[0.73, 1.56]	80%
Exclusion Criteria	Born after 1954	886	209	0.95	[0.74, 1.68]	83%

(continued)

Decision	Option	Successful Universes	Non- convergence	Mean RR	RR Range	% Sig
Exclusion Criteria	Younger than 55	793	229	0.95	[0.76, 1.67]	79%
Outcome	Age event + Death + Event ever + Since	644	193	0.92	[0.73, 1.06]	84%
Outcome	Death + Event ever + Since	656	182	0.92	[0.71, 1.06]	84%
Outcome	Age event + Death + Event ever	707	177	0.93	[0.75, 1.16]	76%
Outcome	Age event + Event ever + Since	617	216	0.93	[0.74, 1.07]	86%
Outcome	Death + Event ever	761	187	0.93	[0.72, 1.13]	76%
Outcome	Event ever	695	266	0.95	[0.73, 1.14]	83%
Outcome	Age event + Event ever	833	266	0.98	[0.77, 1.68]	72%
Adjustment Set	Smoke + Sex	81	19	0.83	[0.74, 1.35]	97%
Adjustment Set	Country + Sex	100	0	0.83	[0.73, 1.12]	95%
Adjustment Set	Sex	100	0	0.83	[0.71, 1.22]	93%
Adjustment Set	Smoke + Edu + Sex	65	35	0.84	[0.75, 1.01]	93%
Adjustment Set	Smoke + Country + Edu + Sex	74	26	0.85	[0.74, 1.16]	94%
Adjustment Set	Smoke + Country + Sex	85	15	0.85	[0.75, 1.03]	90%
Adjustment Set	Edu + Sex	75	25	0.85	[0.76, 1.31]	90%
Adjustment Set	Sex + Phys	100	0	0.85	[0.76, 1.39]	94%
Adjustment Set	Smoke	86	14	0.86	[0.77, 1.08]	98%
Adjustment Set	Smoke + Edu + Sex + Phys	66	34	0.86	[0.77, 1.2]	96%
Adjustment Set	Country + Edu + Sex	73	27	0.86	[0.75, 1.22]	91%
Adjustment Set	Country + Sex + Phys	79	21	0.86	[0.77, 1.01]	97%
Adjustment Set	Edu + Sex + Phys	68	32	0.86	[0.8, 1.02]	96%
Adjustment Set	Smoke + Country + Edu + Sex + Phys	63	37	0.87	[0.78, 1]	93%
Adjustment Set	Smoke + Country + Sex + Phys	72	28	0.87	[0.77, 1.21]	92%
Adjustment Set	Smoke + Sex + Phys	84	16	0.87	[0.8, 1.38]	94%
Adjustment Set	None	100	0	0.87	[0.76, 1.28]	97%
Adjustment Set	Smoke + Country	87	13	0.88	[0.78, 1.26]	96%
Adjustment Set	Country	100	0	0.88	[0.76, 1.24]	98%
Adjustment Set	Country + Edu + Sex + Phys	73	27	0.89	[0.79, 1.25]	91%
Adjustment Set	Edu	78	22	0.89	[0.79, 1.13]	99%
Adjustment Set	Smoke + Country + Edu	66	34	0.90	[0.79, 1.21]	93%
Adjustment Set	Smoke + Edu	67	33	0.90	[0.79, 1.41]	95%
Adjustment Set	Phys	100	0	0.90	[0.81, 1.12]	100%
Adjustment Set	Country + Edu	75	25	0.91	[0.81, 1.21]	93%
Adjustment Set	Smoke + Country + Phys	84	16	0.92	[0.82, 1.1]	94%
Adjustment Set	Smoke + Phys	87	13	0.92	[0.81, 1.25]	93%
Adjustment Set	Country + Phys	100	0	0.92	[0.82, 1.31]	96%
Adjustment Set	Edu + Phys	79	21	0.93	[0.83, 1.24]	98%
Adjustment Set	Smoke + Country + Edu + Phys	69	31	0.94	[0.83, 1.27]	95%
Adjustment Set	Smoke + Edu + Phys	75	25	0.94	[0.84, 1.47]	95%
Adjustment Set	Country + Edu + Phys	72	28	0.94	[0.83, 1.27]	92%
Adjustment Set	Age + Smoke + Sex	74	26	0.96	[0.87, 1.24]	82%
Adjustment Set	Age + Country + Edu + Sex	69	31	0.96	[0.87, 1.02]	63%
Adjustment Set	Age + Edu + Sex	60	40	0.96	[0.88, 1.09]	80%
Adjustment Set	Age + Sex	82	18	0.96	[0.88, 1.03]	88%
Adjustment Set	Age + Smoke + Country + Edu + Sex	62	38	0.97	[0.84, 1.3]	69%
Adjustment Set	Age + Smoke + Edu + Sex	68	32	0.97	[0.89, 1.41]	79%
Adjustment Set	Age + Smoke + Sex + Phys	79	21	0.97	[0.9, 1.33]	77%
Adjustment Set	Age + Country + Sex	77	23	0.97	[0.87, 1.28]	68%
Adjustment Set	Age + Sex + Phys	79	21	0.97	[0.88, 1.56]	78%
Adjustment Set	Age + Smoke + Country + Edu + Sex + Phys	61	39	0.98	[0.88, 1.26]	54%
Adjustment Set	Age + Smoke + Country + Sex	75	25	0.98	[0.84, 1.35]	65%
Adjustment Set	Age + Smoke + Edu + Sex + Phys	62	38	0.98	[0.88, 1.57]	80%
Adjustment Set	Age + Country + Edu + Sex + Phys	80	20	0.98	[0.87, 1.33]	57%
Adjustment Set	Age + Country + Sex + Phys	86	14	0.98	[0.85, 1.33]	57%
Adjustment Set	Age + Edu + Sex + Phys	65	35	0.98	[0.88, 1.33]	76%
Adjustment Set	Age + Smoke + Country + Sex + Phys	72	28	0.99	[0.87, 1.32]	43%
Adjustment Set	Age + Smoke	75	25	1.01	[0.92, 1.04]	37%
Adjustment Set	Age	87	13	1.02	[0.98, 1.42]	33%
Adjustment Set	Age + Smoke + Edu	80	20	1.02	[0.96, 1.13]	27%
Adjustment Set	Age + Smoke + Edu + Phys	61	39	1.03	[0.94, 1.67]	51%
Adjustment Set	Age + Smoke + Country	73	27	1.04	[0.95, 1.42]	44%
Adjustment Set	Age + Smoke + Country + Edu	67	33	1.04	[0.96, 1.2]	60%
Adjustment Set	Age + Smoke + Country + Edu + Phys	66	34	1.04	[0.94, 1.45]	77%

(continued)

Decision	Option	Successful Universes	Non- convergence	Mean RR	RR Range	% Sig
Adjustment Set	Age + Country	87	13	1.04	[0.98, 1.28]	74%
Adjustment Set	Age + Edu	71	29	1.04	[0.99, 1.63]	48%
Adjustment Set	Age + Phys	85	15	1.04	[0.98, 1.34]	60%
Adjustment Set	Age + Smoke + Country + Phys	78	22	1.05	[0.98, 1.4]	82%
Adjustment Set	Age + Smoke + Phys	68	32	1.05	[0.97, 1.68]	58%
Adjustment Set	Age + Country + Edu	63	37	1.05	[0.98, 1.37]	75%
Adjustment Set	Age + Country + Edu + Phys	71	29	1.05	[1, 1.47]	78%
Adjustment Set	Age + Country + Phys	77	23	1.05	[0.99, 1.45]	70%
Adjustment Set	Age + Edu + Phys	70	30	1.05	[1, 1.54]	59%